

LAB MODULE #5: CONTROL ROD CALIBRATION LABORATORY

LABORATORY OBJECTIVES:

1. Learn how to manipulate control rods to increase the net reactivity (ρ_{net}) in the PULSTAR reactor core and put the reactor on a positive period (τ).
2. Calibrate control rod SAFETY #1 by measuring induced periods and calculating corresponding net reactivity values using the Inhour equation.
3. Plot DIFFERENTIAL and INTEGRAL rod worth curves for control rod SAFETY #1 using the measured reactivity data.

EQUIPMENT:

- PULSTAR Reactor.
- Two stopwatches.

PRECONDITIONS:

1. Reactor Daily Startup Checklist completed and signed by Designated Senior Reactor Operator (DSRO).
2. Reactor startup has been authorized by DSRO.
3. Licensed Reactor Operator is on duty at the reactor control console.

EQUATIONS:

$$\tau = 1.44 \cdot dt$$

Where:

τ is the period (seconds)

dt is the power doubling time (seconds).

$$\rho = \frac{\beta_{\text{eff}}}{(1 + \lambda_{\text{eff}} \tau)}$$

Small Reactivity Insertion “One Group” Approximation to Inhour Equation where:

ρ is reactivity (pcm);

$\beta_{\text{eff}} = 730$ pcm; and

λ_{eff} is the effective decay constant = 0.1 (1/sec)

Valid for large values of τ where $\rho \ll \beta_{\text{eff}}$

PROCEDURE:

1. Bring the reactor critical and establish critical gang rod positions at 100 Watts.
2. Place SAFETY #2 and REGULATING rods on gang, and SAFETY #1 off-gang.
3. Insert SAFETY #1 while withdrawing the GANGED RODS to keep reactor power steady at 100 Watts. Insert SAFETY #1 as far as possible while maintaining the reactor at criticality.

4. After establishing the initial SAFETY #1 rod position with the reactor critical at 100 Watts, the following steps will be repeated until the SAFETY #1 rod is fully withdrawn:
 - a) Record initial SAFETY #1 rod position on the data sheet.
 - b) Withdraw the SAFETY #1 rod in one continuous pull until the Log-N period meter reads approximately 30 seconds. Release the rod motion switch. Record final SAFETY #1 rod position on the data sheet.
 - c) Wait until the Linear Power Channel increases and auto-ranges to the 1 kW scale.
 - d) Using two stopwatches, measure the time it takes for reactor power to double as indicated on the Linear channel meter (e.g. 30% to 60% of scale, or 40% to 80% of scale). Record values for doubling times dt (sec) on the data sheet.
 - e) Keeping the SAFETY #1 rod position constant, insert the GANGED RODS and bring the reactor back to criticality at 100 watts.
5. Calculate the average doubling time for each rod withdrawal and calculate the reactor period τ . Using the period value and the Inhour curve, determine the corresponding net reactivity ρ (pcm) for each rod pull. Record values on the data sheet.

ANALYSIS:

1. The reactivity values as determined by the Inhour curve (see attached on page #3) and the Small Reactivity Insertion One-Group Inhour approximation should be in close relative agreement (assuming reactor periods of greater than $\tau = 30$ seconds were utilized).
2. The DIFFERENTIAL ROD WORTH (pcm/inch) for each pull is calculated by taking the reactivity calculated in STEP #5 above and dividing by the change in final minus initial rod heights (Δ inches). This value is then plotted at the midpoint rod height for each pull.
3. The INTEGRAL ROD WORTH (total pcm) is plotted at the endpoint (final) rod position for each withdrawal. It is the sum of the rod worth for each withdrawal plus all previous withdrawals.

